

The Optimal Bourgain Constant in Quantitative Bochner Weak Precompactness

FA/Banach discovery run `fa_banach_001`

Candidate packet, 2026-06-13

Source and open question

The source paper is Jose Rodriguez, “ ε -weakly precompact sets in Banach spaces”, arXiv:2102.01546, submitted 2021-02-02. The open problem addressed here is Question 2.21.

Proof of Theorem 2.11. (ii) follows from (i), Theorem 2.1(i) and Lemma 2.5.

We begin the proof of (i) with the following:

Particular case. Suppose that $f_n \in L_\infty(\mu, X)$ for all $n \in \mathbb{N}$ and that

$$\sup_{n \in \mathbb{N}} \|f_n\|_{L_\infty(\mu, X)} < \infty.$$

Write $Y := L_1([0, 1], X)$. For each $n \in \mathbb{N}$ we define (thanks to Lemma 2.19) $h_n \in L_\infty(\mu, Y)$ by

$$h_n(\omega) := r_n \otimes f_n(\omega) \quad \text{for } \mu\text{-a.e. } \omega \in \Omega.$$

By the assumption on F and Lemma 2.17, we have

$$\text{clust}_{Y^{**}}(h_n(\omega)) \subseteq \varepsilon B_{Y^{**}} \quad \text{for } \mu\text{-a.e. } \omega \in \Omega.$$

Bearing in mind that $\sup_{n \in \mathbb{N}} \|h_n\|_{L_\infty(\mu, Y)} < \infty$, an appeal to Lemma 2.18 ensures that

$$\text{clust}_{L_1(\mu, Y)^{**}}(h_n) \subseteq \varepsilon B_{L_1(\mu, Y)^{**}}.$$

Write $Z := L_1(\mu, X)$ and let $\Phi : L_1(\mu, Y) \rightarrow L_1([0, 1], Z)$ be the natural isometric isomorphism. Then $\Phi(h_n) = r_n \otimes f_n$ for all $n \in \mathbb{N}$ and therefore

$$\text{clust}_{L_1([0, 1], Z)^{**}}(r_n \otimes f_n) \subseteq \varepsilon B_{L_1([0, 1], Z)^{**}}.$$

Therefore, $(f_n)_{n \in \mathbb{N}}$ cannot be an ℓ_1 -sequence with constant $C > \varepsilon$ (by Lemma 2.16), as required.

We now turn to the:

General case. Fix $C > \varepsilon$ and choose any $0 < \eta < C - \varepsilon$. Since $(f_n)_{n \in \mathbb{N}}$ is uniformly integrable, Lemma 2.20 gives a sequence $(A_n)_{n \in \mathbb{N}}$ in Σ such that each $g_n := f_n \chi_{A_n}$ belongs to $L_\infty(\mu, X)$ and:

- $\sup_{n \in \mathbb{N}} \|g_n\|_{L_\infty(\mu, X)} < \infty$,
- $\|f_n - g_n\|_{L_1(\mu, X)} \leq \eta$ for every $n \in \mathbb{N}$.

Define a multi-function $\tilde{F} : \Omega \rightarrow \mathcal{P}(X)$ by $\tilde{F}(\omega) := F(\omega) \cup \{0\}$ for every $\omega \in \Omega$. By the *Particular case*, the sequence $(g_n)_{n \in \mathbb{N}}$ in $S_1(\tilde{F})$ cannot be an ℓ_1 -sequence with constant $C - \eta$. Since $\|f_n - g_n\|_{L_1(\mu, X)} \leq \eta$ for all $n \in \mathbb{N}$, we conclude that $(f_n)_{n \in \mathbb{N}}$ cannot be an ℓ_1 -sequence with constant C . The proof is finished. $\square$

Question 2.21. *Is constant 2 optimal in Theorem 2.11?*

The concept of $\delta\mathcal{C}$ -sets in Lebesgue-Bochner spaces (which corresponds to the case $\varepsilon = 0$ of the following definition) appeared first in [8] and was discussed further in [6].

The question asks whether the constant 2 in Rodriguez's Theorem 2.11 is optimal. In the notation of that theorem, (Ω, Σ, μ) is a probability space, X is a Banach space, $F : \Omega \rightarrow \mathcal{P}(X)$ has

$F(\omega)$ ε -weakly precompact for almost every ω , and

$$S_1(F) = \{f \in L_1(\mu, X) : f(\omega) \in F(\omega) \text{ for a.e. } \omega\}.$$

The source theorem proves that every uniformly integrable subset of $S_1(F)$ is 2ε -weakly precompact.

Result

Theorem 1. *In Theorem 2.11 of the source paper the constant 2 can be replaced by 1. More precisely, if $F(\omega)$ is ε -weakly precompact for almost every ω , then:*

- (i) *no uniformly integrable sequence in $S_1(F)$ is an ℓ_1 -sequence with constant $C > \varepsilon/2$;*
- (ii) *every uniformly integrable subset of $S_1(F)$ is ε -weakly precompact in $L_1(\mu, X)$.*

Consequently, for every ε -weakly precompact set $M \subseteq X$, the set $L(M)$ is ε -weakly precompact in $L_1(\mu, X)$.

Theorem 2. *The constant 1 in Theorem 1 is optimal. In particular, Question 2.21 has a negative answer: the constant 2 is not optimal, and the best universal constant is 1.*

Definitions used

For a bounded sequence (x_n) in a Banach space X , let

$$\delta(x_n) = \text{diam clust}_{X^{**}}(x_n) = \sup_{x^* \in B_{X^*}} \left(\limsup_n x^*(x_n) - \liminf_n x^*(x_n) \right).$$

A bounded set $M \subseteq X$ is ε -weakly precompact if every sequence in M has a subsequence (x_{n_k}) with $\delta(x_{n_k}) \leq \varepsilon$. An ℓ_1 -sequence with constant C is a bounded sequence (x_n) satisfying

$$\left\| \sum_{j=1}^m a_j x_j \right\| \geq C \sum_{j=1}^m |a_j|$$

for all finite scalar families.

The midpoint lemma

The source proof of Lemma 2.17 uses Rodriguez's Lemma 2.14 with the centre $\liminf h_n$. The factor 2 is lost exactly there. The following variant centres at the midpoint of the scalar cluster interval.

Lemma 1. *Let (Ω, Σ, μ) be a probability space, let (h_n) be a bounded sequence in $L_\infty(\mu)$, and suppose*

$$\limsup_n h_n - \liminf_n h_n \leq \varepsilon \quad \text{a.e.}$$

Put

$$m = \frac{1}{2} \left(\limsup_n h_n + \liminf_n h_n \right) \in L_\infty(\mu).$$

Then

$$\limsup_n \int_\Omega |h_n - m| d\mu \leq \frac{\varepsilon}{2}.$$

Proof. Let $g_n = |h_n - m|$. The sequence (g_n) is bounded in $L_\infty(\mu)$. Pointwise,

$$\limsup_n g_n \leq \frac{1}{2} \left(\limsup_n h_n - \liminf_n h_n \right) \leq \frac{\varepsilon}{2}.$$

Rodriguez's Lemma 2.13, the reverse Fatou inequality for bounded L_∞ -sequences on a probability space, gives

$$\limsup_n \int_\Omega g_n d\mu \leq \int_\Omega \limsup_n g_n d\mu \leq \frac{\varepsilon}{2}.$$

□

Improved Rademacher lemma

Let (r_n) be the Rademacher functions on $[0, 1]$.

Lemma 2. *Let Z be a Banach space, let $M \subseteq Z$ be ε -weakly precompact, and let (z_n) be a sequence in M . Then*

$$\text{clust}_{L_1([0,1],Z)^{**}}(r_n \otimes z_n) \subseteq \frac{\varepsilon}{2} B_{L_1([0,1],Z)^{**}}.$$

Proof. The proof is Rodriguez's Lemma 2.17 with Lemma 1 substituted for Lemma 2.14. Let F be a weak-star cluster point of $(r_n \otimes z_n)$ and fix $\varphi \in B_{L_1([0,1],Z)^*}$. Represent φ by a weak-star scalarly measurable function $[0, 1] \rightarrow Z^*$ with $\|\varphi(t)\| \leq 1$ a.e. Choose a subsequence (n_k) such that

$$\langle r_{n_k} \otimes z_{n_k}, \varphi \rangle \rightarrow \langle F, \varphi \rangle.$$

Passing to a further subsequence, using ε -weak precompactness of M , we may suppose $\delta(z_{n_k}) \leq \varepsilon$.

Set

$$a_k(t) = \langle z_{n_k}, \varphi(t) \rangle.$$

Then (a_k) is bounded in $L_\infty[0, 1]$, and for almost every t ,

$$\limsup_k a_k(t) - \liminf_k a_k(t) \leq \delta(z_{n_k}) \leq \varepsilon.$$

Let

$$m(t) = \frac{1}{2} \left(\limsup_k a_k(t) + \liminf_k a_k(t) \right).$$

By Lemma 1,

$$\limsup_k \int_0^1 |a_k(t) - m(t)| dt \leq \frac{\varepsilon}{2}.$$

Since (r_{n_k}) is weakly null in $L_1[0, 1]$ and $m \in L_\infty[0, 1]$,

$$\int_0^1 r_{n_k}(t) m(t) dt \rightarrow 0.$$

Thus

$$|\langle r_{n_k} \otimes z_{n_k}, \varphi \rangle| \leq \left| \int_0^1 r_{n_k}(t) m(t) dt \right| + \int_0^1 |a_k(t) - m(t)| dt,$$

and taking $\limsup_k$ yields $|\langle F, \varphi \rangle| \leq \varepsilon/2$. Since φ was arbitrary in the dual unit ball, $\|F\| \leq \varepsilon/2$. □

Proof of Theorem 1

First assume the particular bounded case used in Rodriguez's proof: each $f_n \in L_\infty(\mu, X)$ and $\sup_n \|f_n\|_{L_\infty(\mu, X)} < \infty$. Let $Y = L_1([0, 1], X)$ and define

$$h_n(\omega) = r_n \otimes f_n(\omega) \in Y.$$

For almost every ω , Lemma 2 applied to the sequence $(f_n(\omega))$ in $F(\omega)$ gives

$$\text{clust}_{Y^{**}}(h_n(\omega)) \subseteq \frac{\varepsilon}{2} B_{Y^{**}}.$$

Rodriguez's Lemma 2.18 then gives

$$\text{clust}_{L_1(\mu, Y)^{**}}(h_n) \subseteq \frac{\varepsilon}{2} B_{L_1(\mu, Y)^{**}}.$$

Under the natural isometric isomorphism $L_1(\mu, L_1([0, 1], X)) \cong L_1([0, 1], L_1(\mu, X))$, this says

$$\text{clust}_{L_1([0, 1], L_1(\mu, X))^{**}}(r_n \otimes f_n) \subseteq \frac{\varepsilon}{2} B.$$

Rodriguez's Lemma 2.16 states that if (f_n) is an ℓ_1 -sequence with constant C , then $(r_n \otimes f_n)$ is an ℓ_1 -sequence with the same constant C , and in particular its bidual cluster set is not contained in ηB for any $\eta < C$. Therefore the bounded particular case cannot contain an ℓ_1 -sequence with constant $C > \varepsilon/2$.

Now let (f_n) be any uniformly integrable sequence in $S_1(F)$ and suppose, for contradiction, that it is an ℓ_1 -sequence with constant $C > \varepsilon/2$. Choose $0 < \eta < C - \varepsilon/2$. Rodriguez's truncation Lemma 2.19 gives measurable sets A_n such that $g_n = f_n \chi_{A_n}$ is uniformly L_∞ -bounded and

$$\|f_n - g_n\|_{L_1(\mu, X)} \leq \eta \quad (n \in \mathbb{N}).$$

The sequence (g_n) belongs to $S_1(\tilde{F})$ for $\tilde{F}(\omega) = F(\omega) \cup \{0\}$, whose values are still ε -weakly precompact. Moreover (g_n) is an ℓ_1 -sequence with constant at least $C - \eta > \varepsilon/2$. This contradicts the bounded case. Hence part (i) follows.

For part (ii), let $W \subseteq S_1(F)$ be uniformly integrable. Fix $\varepsilon' > \varepsilon$. If W were not ε' -weakly precompact, Theorem 2.1(i) of the source paper would give an ℓ_1 -sequence in W with constant $\varepsilon'/2 > \varepsilon/2$, contradicting part (i). Thus W is ε' -weakly precompact for every $\varepsilon' > \varepsilon$, and Rodriguez's Lemma 2.5 implies that W is ε -weakly precompact.

Proof of optimality

Let $X = \ell_1$ and let $M = \{\alpha e_n : n \in \mathbb{N}\}$ for some $\alpha > 0$. In the one-point probability space, $L_1(\mu, X)$ identifies with X and $L(M) = M$.

The set M is 2α -weakly precompact. Indeed, every sequence in M has either a constant subsequence, with $\delta = 0$, or a subsequence of distinct basis vectors. For a distinct basis subsequence, the diameter is exactly 2α : it is at most 2α by norm diameter, and it is at least 2α by evaluating against a bounded scalar sequence in ℓ_∞ which alternates between 1 and -1 along the chosen indices.

Thus M is not η -weakly precompact for any $\eta < 2\alpha$. If a universal constant $K < 1$ replaced the constant 1 in Theorem 1, applying it with $\varepsilon = 2\alpha$ would force $L(M) = M$ to be $2K\alpha$ -weakly precompact, contradicting $2K\alpha < 2\alpha$. Hence no constant smaller than 1 is possible. The same example shows the threshold $C > \varepsilon/2$ in part (i) is sharp.

Novelty and literature check

Before promotion, bounded searches were run on 2026-06-13 for exact and close phrases involving:

- Question 2.21, 2102.01546, and `epsilon-weakly precompact`;
- Theorem 2.11, constant 2, and Bourgain;
- Bourgain–Maurey–Pisier, weakly conditionally compact sets in $L_1 X$, and Bourgain’s averaging result for ℓ_1 -sequences.

The searches found Rodriguez’s arXiv paper and background around Bourgain, Maurey–Pisier, Behrends, Gasparis, and Kalenda–Pfitzner–Spurny, but no separate later source explicitly resolving Question 2.21 or advertising the constant-one improvement. This is a bounded novelty check, not a claim of exhaustive bibliographic certainty.

Verification notes and human-review focus

The proof is short and the review focus should be correspondingly narrow:

1. Check Lemma 1: it uses only the source paper’s reverse Fatou Lemma 2.13 for bounded L_∞ sequences.
2. Check that the midpoint function m in Lemma 2 is measurable and in L_∞ ; this follows because it is built from $\limsup$ and $\liminf$ of bounded measurable scalar functions.
3. Check that the passage from the pointwise improved Rademacher lemma to the global Bochner statement is exactly the source proof of Theorem 2.11, with ε replaced by $\varepsilon/2$ up to the final Behrends endpoint argument.
4. Check the one-point ℓ_1 example for optimality.

Current verdict: likely valid candidate full solution, pending human review.

Human Review: Optimal Bourgain constant

Original automatic proof: `solution_packet.pdf`

Checked date: 14 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proposed proof appears to give a correct improvement of the constant in Rodriguez's Theorem 2.11 from 2 to 1. The key point is identifying exactly where the factor ε is lost in the original proof and replacing that estimate by a sharper midpoint estimate.

In Rodriguez's Lemma 2.17, one considers scalar functions of the form

$$a_k(t) = \langle z_{n_k}, \varphi(t) \rangle$$

whose pointwise oscillation is bounded by ε :

$$\limsup_k a_k(t) - \liminf_k a_k(t) \leq \varepsilon \quad \text{for a.e. } t.$$

The original proof centers $a_k(t)$ at the endpoint

$$\liminf_k a_k(t).$$

This controls the error by the full oscillation, and hence gives an ε -bound.

The corrected argument centers instead at the midpoint

$$m(t) = \frac{1}{2} \left(\limsup_k a_k(t) + \liminf_k a_k(t) \right).$$

Then

$$\limsup_k |a_k(t) - m(t)| \leq \frac{1}{2} \left(\limsup_k a_k(t) - \liminf_k a_k(t) \right) \leq \frac{\varepsilon}{2}$$

for almost every t . Applying Rodriguez's reverse Fatou lemma, Lemma 2.13 of the source paper, gives

$$\limsup_k \int_0^1 |a_k(t) - m(t)| dt \leq \frac{\varepsilon}{2}.$$

Improved Lemma 2.17

With this midpoint replacement, the proof of Rodriguez's Lemma 2.17 gives

$$\text{clust}_{L_1([0,1],Z)^{**}}(r_n \otimes z_n) \subseteq \frac{\varepsilon}{2} B_{L_1([0,1],Z)^{**}}$$

instead of the original εB -bound.

The last step in the estimate is unchanged. One writes

$$a_k(t) = m(t) + (a_k(t) - m(t)).$$

Then

$$|\langle r_{n_k} \otimes z_{n_k}, \varphi \rangle| \leq \left| \int_0^1 r_{n_k}(t) m(t) dt \right| + \int_0^1 |a_k(t) - m(t)| dt.$$

The first term tends to zero because the Rademacher sequence is weakly null in $L_1[0,1]$ and $m \in L_\infty[0,1]$. The second term is bounded by $\varepsilon/2$ in limsup by the midpoint-Fatou estimate above.

Propagation Through the Original Proof

Once this sharpened version of Lemma 2.17 is available, the rest of the proof is essentially Rodriguez's original proof of Theorem 2.11. The pointwise cluster estimate is passed to the Bochner L_1 -space using Rodriguez's Lemma 2.18. Then Lemma 2.16 excludes uniformly integrable ℓ_1 -sequences with constant $C > \varepsilon/2$. The general uniformly integrable case is handled by the same truncation argument as in the source paper.

Finally, the endpoint passage from all $\varepsilon' > \varepsilon$ back to ε uses Rodriguez's existing closedness lemma for ε -weak precompactness. The one-point probability space example with the scaled unit vector basis of ℓ_1 shows that the constant 1 cannot be improved.

Review Outcome

Archive this packet as an apparently correct full solution to Question 2.21: the constant 2 is not optimal, and the optimal universal constant is 1. The mathematical content is concentrated in the midpoint correction to the scalar oscillation estimate; after that correction, the argument follows the architecture of the original paper.